CO2xN: Porewater Ammonium

September 2025 Samples

2026-01-14

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##Run Information

```r
cat("Run Information: ") #lets you know what section you're in
```

```
## Run Information:
```

```r
#set the run date (MM/DD/YYYY) & user name
  run_date <- "01/14/2026"
  sample_year <- "2025"
  sample_month <- "July"
  user <- "Brenden Blakley"

#identify the files you want to read in
  #read in as a list to accommodate multiple runs in a month
  NH3_files <- c("Raw Data/LTREB_2M_Sept_2025_Run1.csv",
                 "Raw Data/LTREB_2M_Sept_2025_Run2.csv")

# Define the file path for QAQC log file - NO Need to change just check year
  file_path <- "Raw Data/SEAL_CO2xN_QAQC_Log_2025.csv"
  final_path <- "Processed Data/CO2xN_Porewater_NH4_202509.csv"

#record any notes about the run or anything other info here:
  run_notes <- ""

  #Collection_Dates = "Raw Data/Sample_Collection_Dates_2024.csv"

  cat(run_notes)
```

##Setup

##Read in metadata and create similar sample IDs for matching to samples

0.1 Import Data & Clean

0.2 Assessing standard Curves

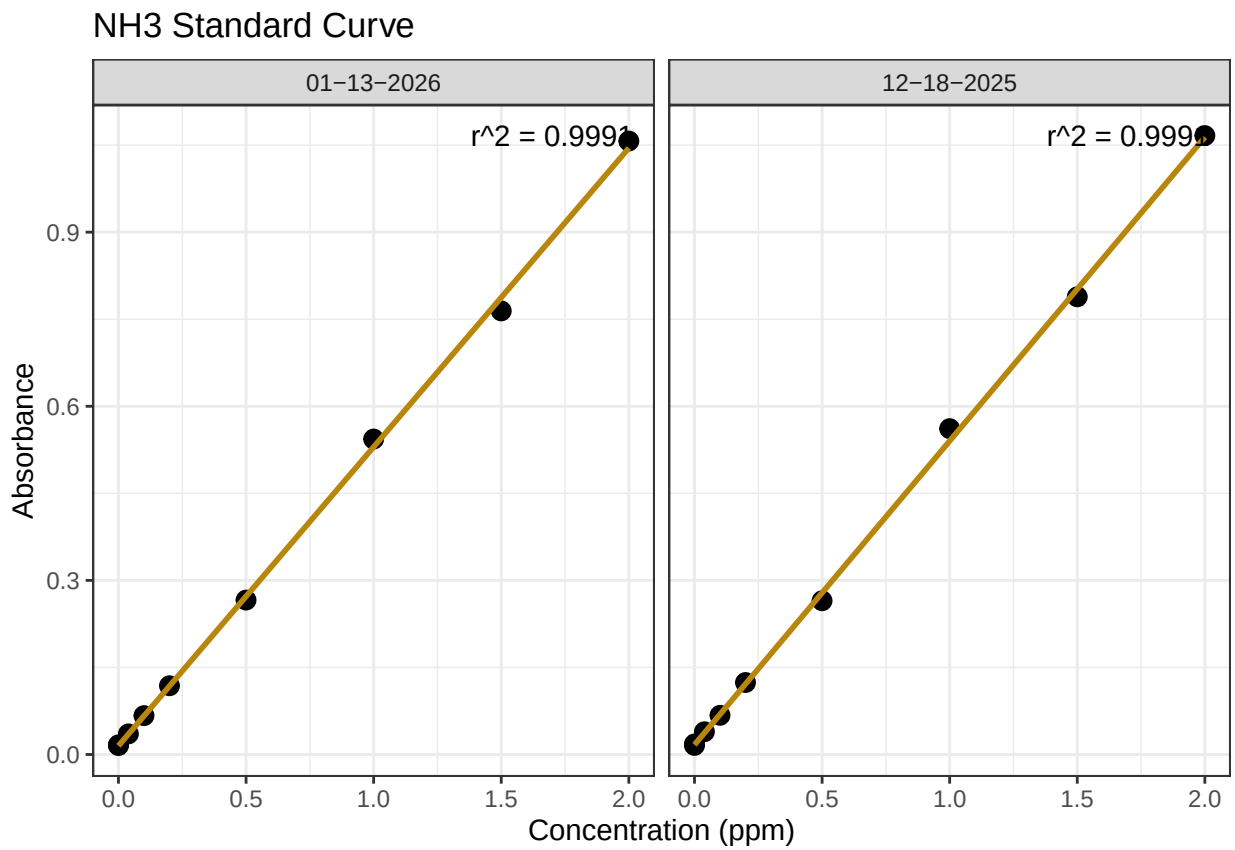
#Pull out standards data

```
## Assess Standard Curves
```

#Plot standards data

```
## Assess Standard Curves
```

```
## `geom_smooth()` using formula = 'y ~ x'
```




```
## [1] "NH3 Curve r2 GOOD - PROCEED"
```

```
## [1] "QAQC log file exists and has been read into the code."
```

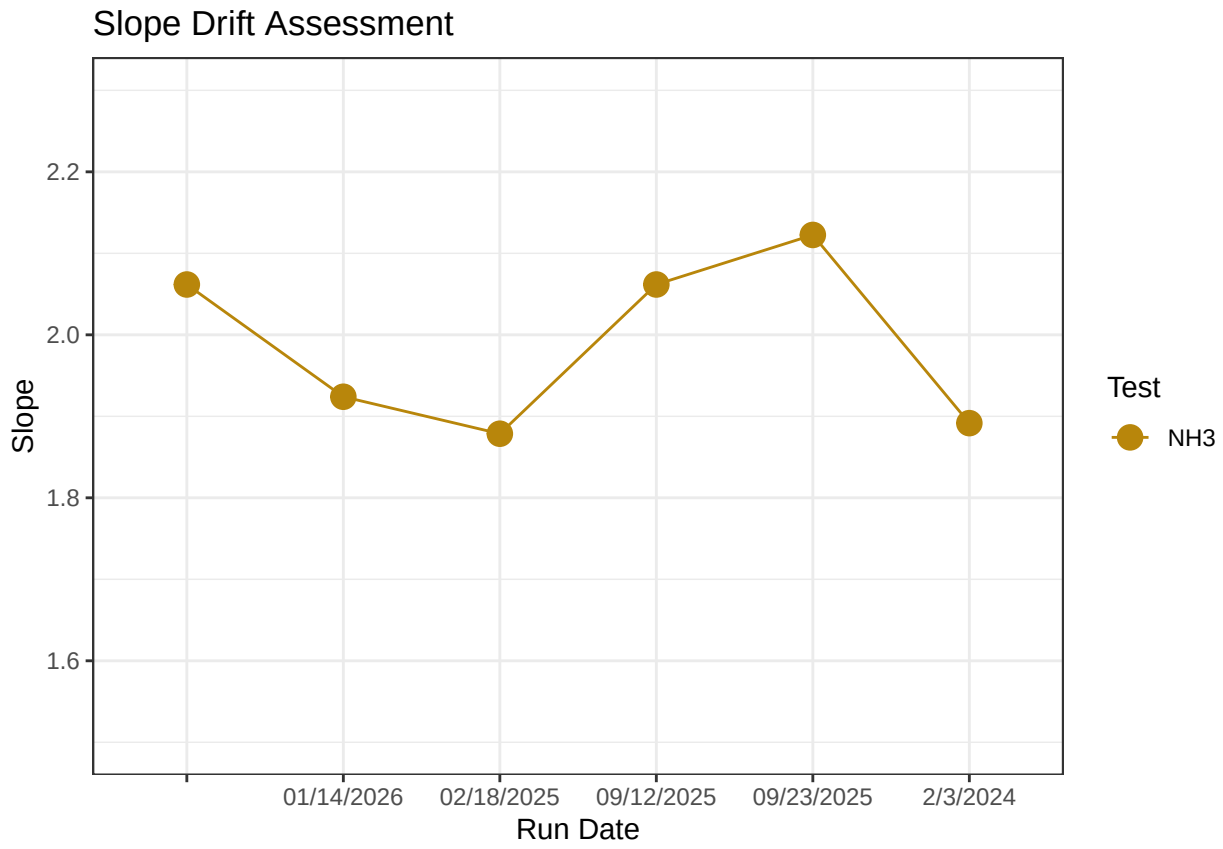


Table 1: Average Slope by Analyte

Test	avg_slope
NH3	1.99

0.3 Dilution Corrections - ensure the latest dilution is kept

```
## Dilution Corrections

## Duplicated samples: , L1-40, L1-80, L4-80, L5-80, L9-80, L10-80, L11-80, L12-80, L14-20, L18-80, L20

## Duplicated samples with NO dilution present (possible input error):
```

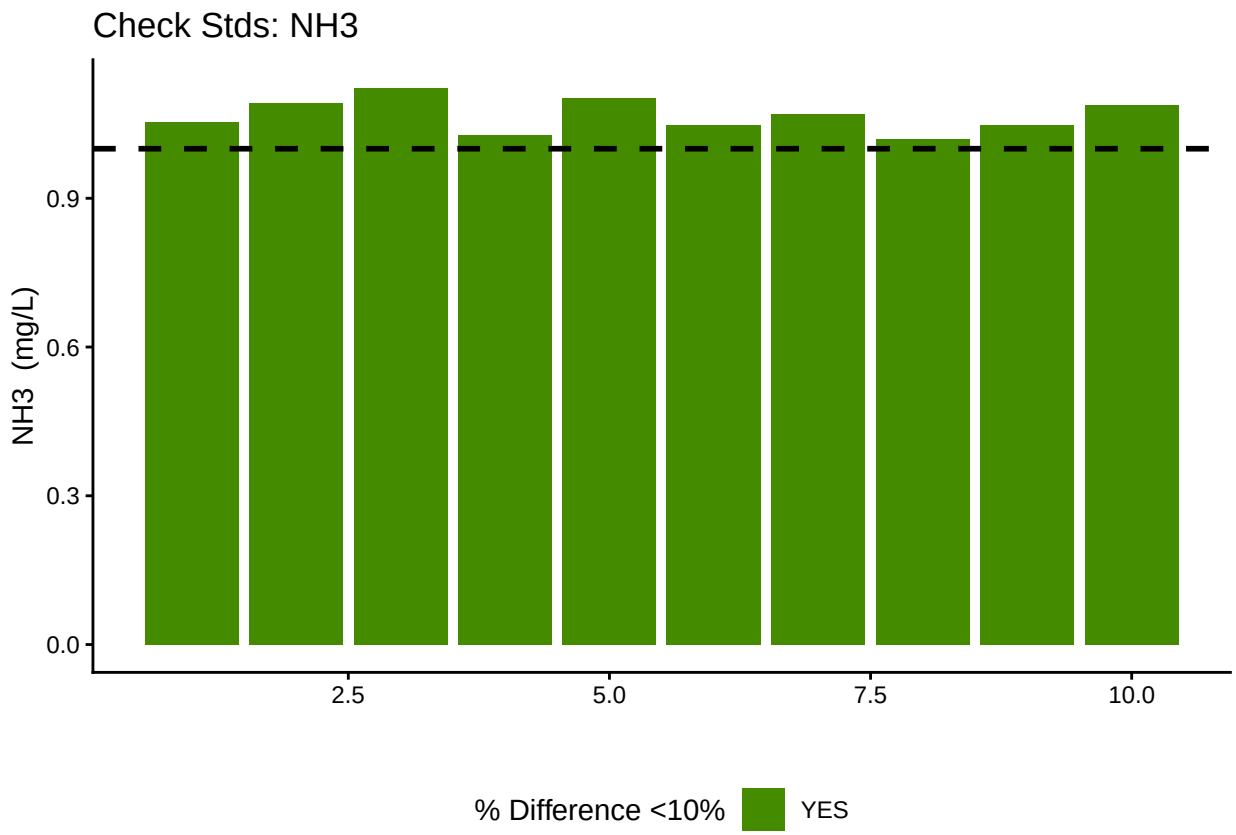
0.4 Performance Check

```
## [1] "NH3 pe Check has a % Difference <25% - PROCEED"

## Run mean = 1.155316

## Expected  = 1.034
```

0.5 Analyze the Check Standards

```
## Analyze Check Standards
```

```
## [1] "NH3 Check Standard RSD within Range - PROCEED"
```




```
## [1] ">60% of NH3 Check Standards are within range of expected concentration - PROCEED"
```

0.6 Analyze Blanks

```
## Assess Blanks
```

```
## [1] ">60% of NH3 Blanks are below the lower 25% quartile of samples - PROCEED"
```

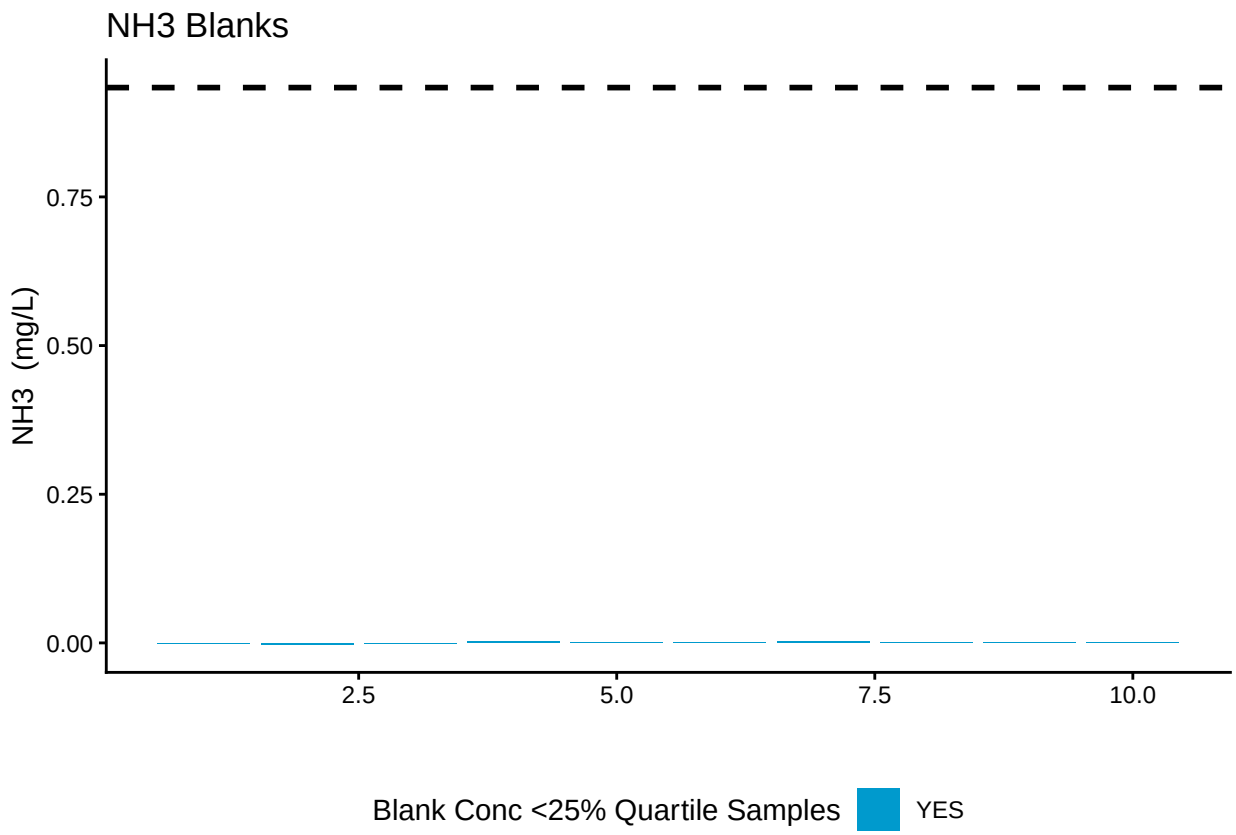


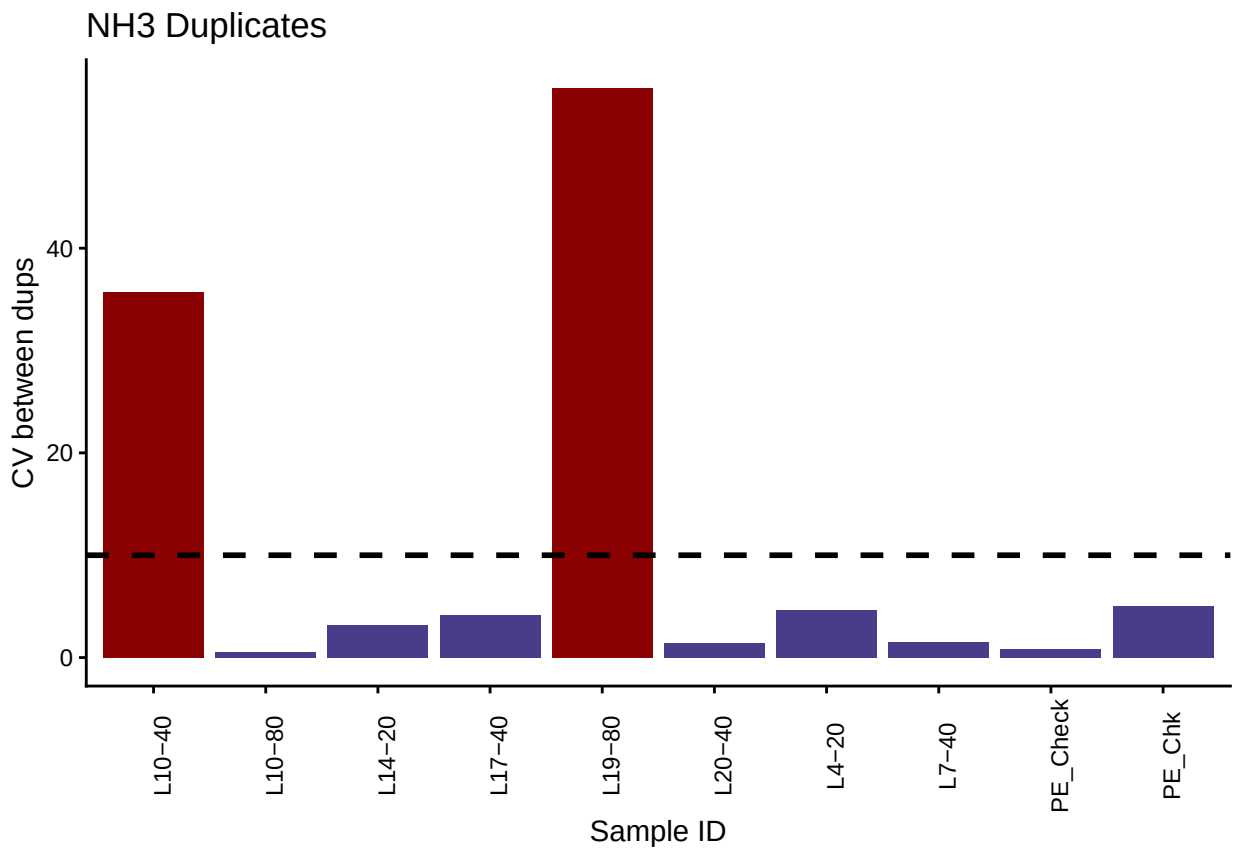
Table 2: Mean Concentration of Blanks

Test	Blank_Mean_Conc
NH3	3e-04

0.7 Analyze Duplicates

```
## Analyze Duplicates
```

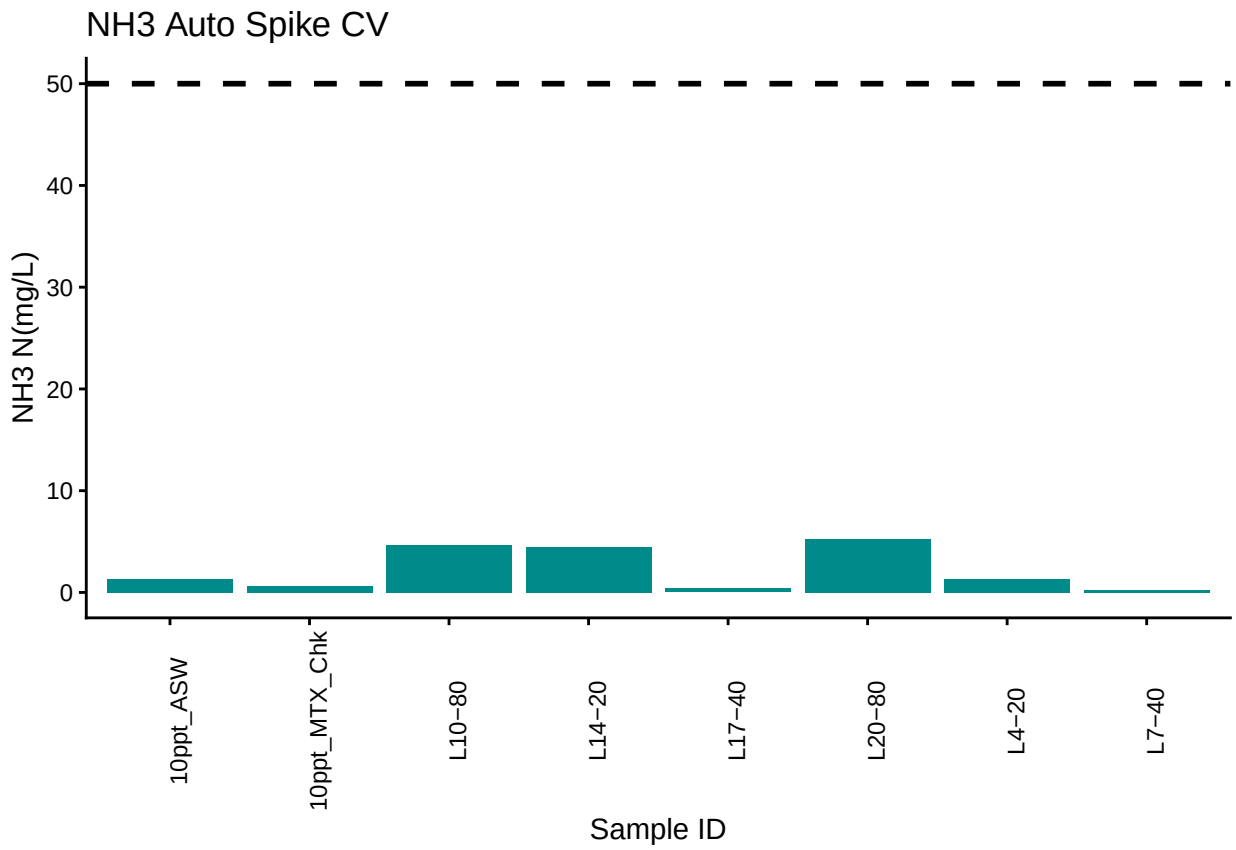
```
## [1] ">60% of NH3 Duplicates have a CV <10% - PROCEED"
```



0.8 Spikes

```
## [1] ">60% of Spikes have a CV <50% - PROCEED"
```

```
## Warning: Using ‘size‘ aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use ‘linewidth‘ instead.
## This warning is displayed once per session.
## Call ‘lifecycle::last_lifecycle_warnings()‘ to see where this warning was
## generated.
```



0.9 Matrix Effects

```
## [1] "NO NH3 Matrix Effect, PROCEED"
```

0.10 Unit Converted Data Column Added (mg/L to uM)

0.11 Sample Flagging - Within range of standard curve

```
## Sample Flagging
```

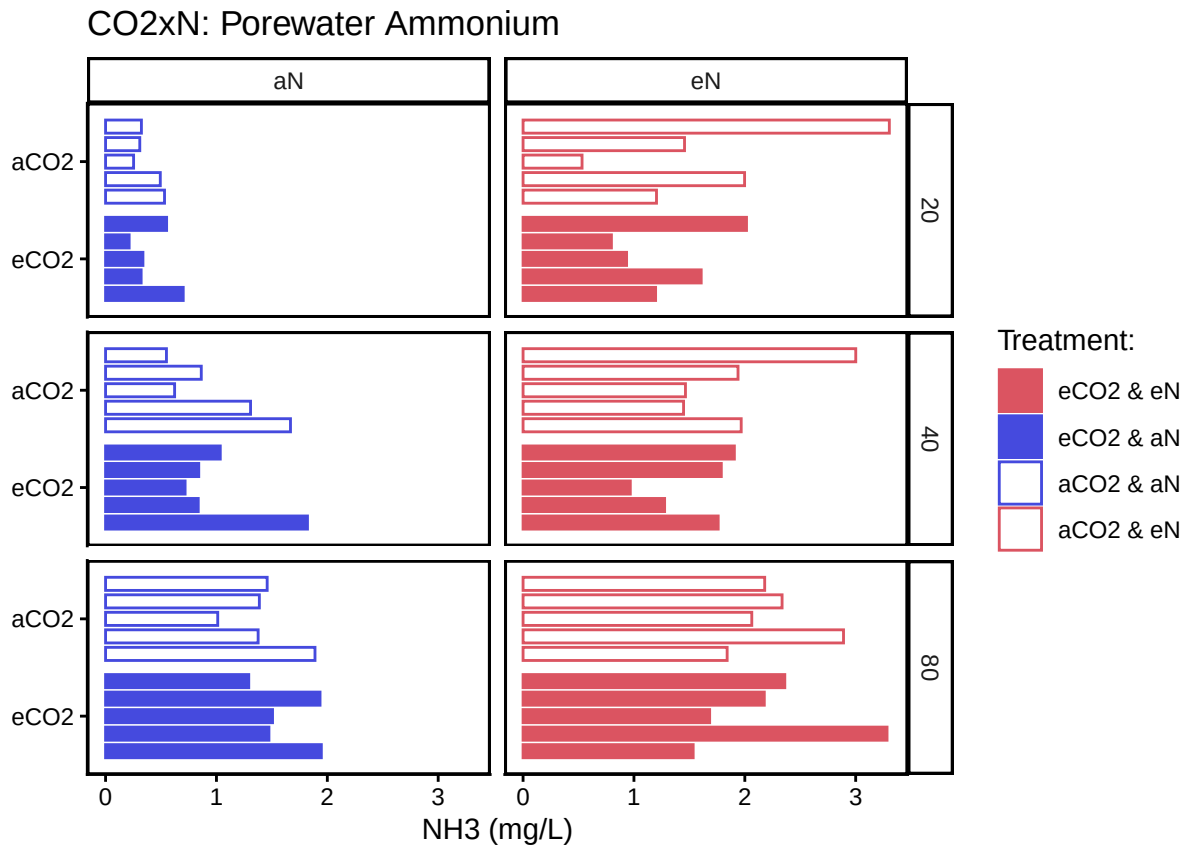
0.12 Merge samples with Metadata & check all samples are present

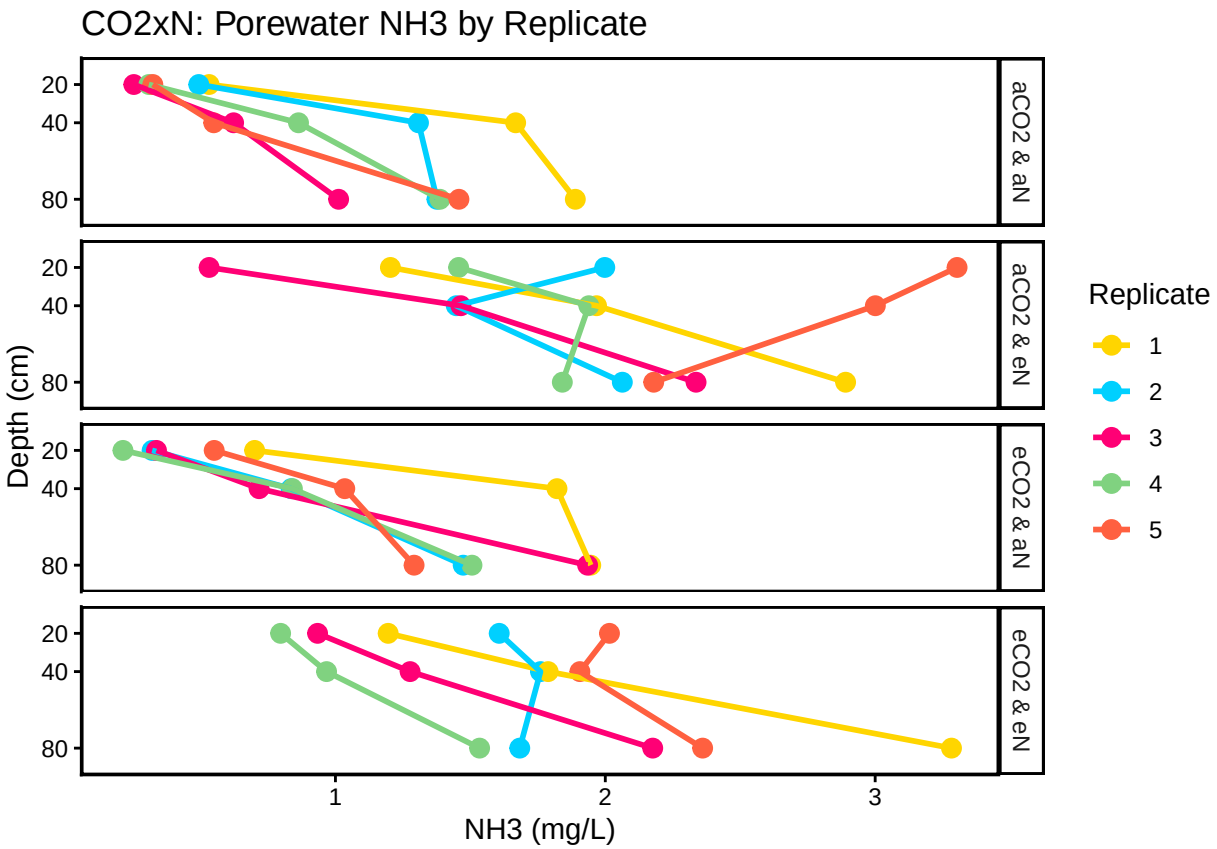
```
## Merge Samples with Metadata
```

```
## All sample IDs are present in data
```

0.13 Organize Data

0.14 Visualize Data





0.15 Export Processed Data

#end